

Bayes Update Cards Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Medium	ID	bayes-update-cards

Big idea: New evidence updates prior probability; it does not replace the background rate.

One-Minute Teacher Brief

Students use natural-frequency cards to see how prior probability and new evidence combine in Bayesian updating.

Student Prompt

Inspect Bayes Update Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Bayes Update Cards Stimulus A	Student-facing stimulus 1: Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Bayes Update Cards Stimulus B	Student-facing stimulus 2: Notation contrast card: $P(H +)$ is not the same as $P(+ H)$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Bayes Update Cards Reveal / Twist	Student-facing stimulus 3: Notation contrast card: $P(H +)$ is not the same as $P(+ H)$. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — New evidence updates prior probability; it does not replace the background rate.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Bayes Update Cards Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Bayes Update Cards Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	How does notation shape mathematical knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how new evidence updates prior probability; it does not replace the background rate.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: How does notation shape mathematical knowledge?

Setup Checklist

- Prepare counters in two colours: true positives and false positives.
- Hide notation at first; reveal $P(H | +)$ only after students build the frequency table.

Example Stimuli or Materials

- Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases.
- Notation contrast card: $P(H | +)$ is not the same as $P(+ | H)$.

Resources To Use

- Resource pack: <resources/bayes-update-cards/index.html>
- Card deck CSV: <resources/bayes-update-cards/cards.csv>
- Visual prompt: <resources/bayes-update-cards/visual-prompt.svg>
- Natural-frequency grid with true positives, false positives, false negatives, and true negatives.
- Risk communication comparison card: words, percentage, frequency, visual grid, and notation.

Run Sequence

1. Give students a fictional condition, base rate, and test accuracy in words.
2. Ask for an intuitive estimate after a positive result.
3. Build the case using natural frequencies, separating true positives from false positives.
4. Introduce the update calculation $P(H | +) = 9 / (9 + 99) = 1 / 12 \approx 8.3\%$.
5. Compare how the same information feels in words, percentages, frequencies, and notation.

Debrief Moves

- Knowledge question: How does mathematical notation change what is visible in a knowledge claim?
- Why do many knowers confuse $P(H | +)$ with $P(+ | H)$?

- What responsibilities come with communicating risk to non-specialists?

Teacher Quick Script

- The notation is not decoration; it tells us which direction the probability is asking about.
- Point to the numerator and denominator before you calculate.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...